CocoConnect

System Design Document – Buyer Interface

Screens, user interactions, navigation, wireframes and design system

Client: Coconut Growers Association of Sri Lanka · Group 38

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1. Purpose and Scope

This document describes the design of the buyer side of the CocoConnect mobile app: the screens a buyer uses, what each screen must contain, how the buyer interacts with it, how the screens connect, and the visual design system. It covers the Product Marketplace and Profile Management parts of the buyer interface and links each screen to the functional requirements in SRS Version 3.

2. Design Principles

Principle	Why	How it shows in the design
Simple and familiar	Many users have limited digital experience (SRS 2.6, NFR-12)	Large cards, one main action per screen, short labels, icons with text
Three languages	Stakeholders use English, Sinhala and Tamil	EN | SI | TA switch in the header of every main screen
Build trust	Buyers and sellers do not trust each other today	Verified badge, trust score and location on every listing; offers kept in a visible history
Transparent prices	Prices are negotiated manually today	Price per nut shown on every listing; offers shown as clear price cards
Location matters	Buyers want supply close to them	Nearby listings with distance; filter by distance
Consistent navigation	NFR-13: clear and consistent navigation	Same bottom bar on every main screen; back arrow on every detail screen

3. Buyer Goals

- Find coconut supply of the right type, quantity and date, close to them.
- Post a request so estate owners can offer to supply it.
- Compare estates by price, trust and distance.
- Agree a price with an estate owner and confirm the order.
- Track orders and keep their profile up to date.

4. Screen Requirements and Interactions

4.1 Buying Categories (buyer dashboard)

Item	Design
Purpose	The buyer's home screen: choose what to buy
Elements	App name, language switch, title 'Buying Categories', cards for Coconuts, Vegetables, Livestock and Other Goods (icon, name, short description), bottom navigation
Interactions	Tap a category card to open it; tap a bottom bar item to switch section
Requirements	FR-10 browse products; FR-14 product categories; NFR-15 room for future products

4.2 Coconut Categories

Item	Design
Purpose	Choose the coconut form before seeing listings
Elements	Cards for Raw (husk on), Husked and Dehusked coconuts, each with a photo, name and short description
Interactions	Tap a type to open the marketplace filtered to that type
Requirements	FR-13 selling method and category; stakeholder finding that coconuts are sold with or without the husk

4.3 Product Marketplace

Item	Design
Purpose	Request coconuts and browse nearby estate listings on one screen

Item	Design
Elements	Request form: number of nuts, required date (date picker), bidding price per nut, Submit Request button. Below: 'Nearby Estate Owner Listings' with distance, Filter button, listing cards (estate name and owner, verified price badge, trust score, price per nut, quantity, available date)
Interactions	Fill and submit the request; open the filter to sort by price or distance; tap a listing card to open it
Validation	Quantity and price must be above 0; date cannot be in the past; clear message beside a wrong field
Requirements	FR-10 browse; FR-11 search and filter by category, location and price; FR-12 product details; FR-13 price, quantity and location shown

4.4 Estate Owner Request (listing detail)

Item	Design
Purpose	Show one estate's offer before contacting the owner
Elements	Back arrow, title, estate photo, estate and owner name, location, product (e.g. raw coconuts), quantity, date, Negotiate button
Interactions	Tap Negotiate to start a conversation with the estate owner
Requirements	FR-12 view product details; FR-15 contact sellers through the platform

4.5 Negotiation

Item	Design
Purpose	Agree a price with the estate owner
Elements	Estate name in the header, status pill, Current Offer card (product, quantity, price per nut), chat messages with times, offer cards inside the chat, Reject and Accept Deal buttons
Interactions	Send a message or a price offer; accept or reject an offer from the other side; accepting confirms the order
Requirements	FR-15 contact sellers; FR-40 notify sellers when buyers contact them; business rule: price agreed before work starts

4.6 Orders

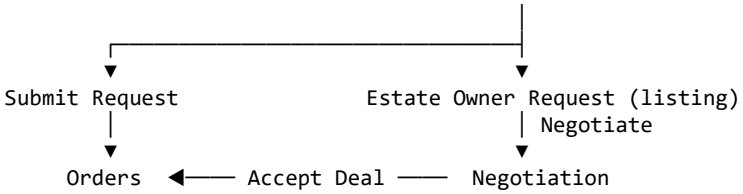
Item	Design
Purpose	Track requests and orders
Elements	Tabs for pending requests, active orders and completed orders; each card shows estate, quantity, price, date and status
Interactions	Tap an order to see details or reopen the negotiation
Requirements	FR-12 view details; order status for transparency

4.7 Profile Management

Item	Design
Purpose	View and manage the buyer's own account
Elements	Profile picture, name, role, phone, location, verification status, rating; options: Edit Profile, Change Password, Language, Log Out
Edit Profile	Form with name, phone, location (pick on map) and profile picture; Save button; same validation style as the request form
Interactions	Edit and save details; change language; log out with a confirmation
Requirements	FR-04 update profile; FR-05 profile picture; FR-07 log out; FR-37 display ratings

5. Navigation Flow

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Login —▶ Buying Categories —▶ Coconut Categories —▶ Marketplace
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```
Bottom bar on every main screen:  Dashboard · Orders · Profile
Profile → Edit Profile / Change Password / Language / Log Out
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- Every main screen has the same bottom bar, so the buyer is never more than one tap from Dashboard, Orders or Profile.
- Detail screens (listing, negotiation, edit profile) have a back arrow and no bottom bar, keeping attention on the task.

6. Key User Flows

6.1 Request raw coconuts

1. Buyer opens Coconuts, then Raw Coconuts.
2. Enters number of nuts, required date and bidding price.
3. Taps Submit Request; a confirmation appears and the request shows in Orders.
4. Estate owners nearby can see the request and send offers.

6.2 Buy from a listing

5. Buyer browses nearby listings and filters by price or distance.
6. Opens a listing and taps Negotiate.
7. Sends or receives price offers in the negotiation screen.
8. Taps Accept Deal on an acceptable offer; the order appears in Orders.

6.3 Update profile

9. Buyer opens Profile, then Edit Profile.
10. Changes details or picture and taps Save; a confirmation appears.

7. Wireframes

The buyer wireframes below show the dashboard, marketplace, listing detail and negotiation screens.

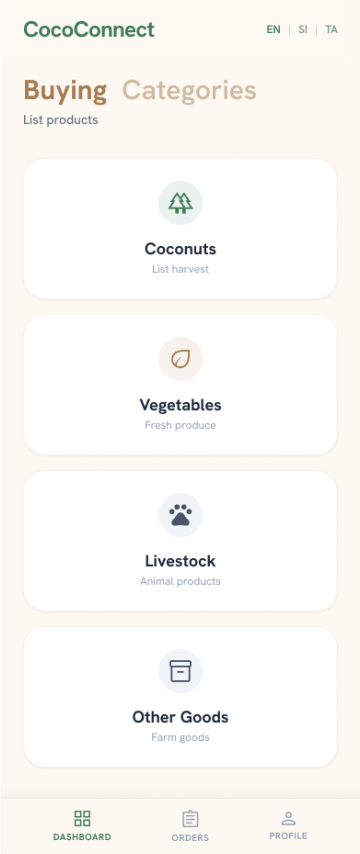


Figure 1: Buyer dashboard (Buying Categories)

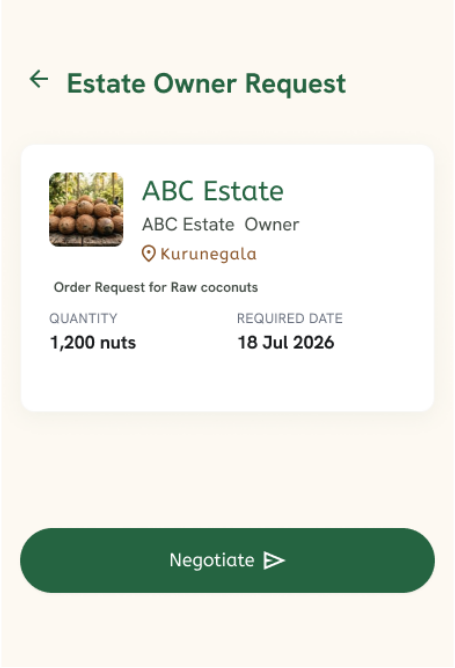


Figure 2: Estate Owner Request (listing detail)

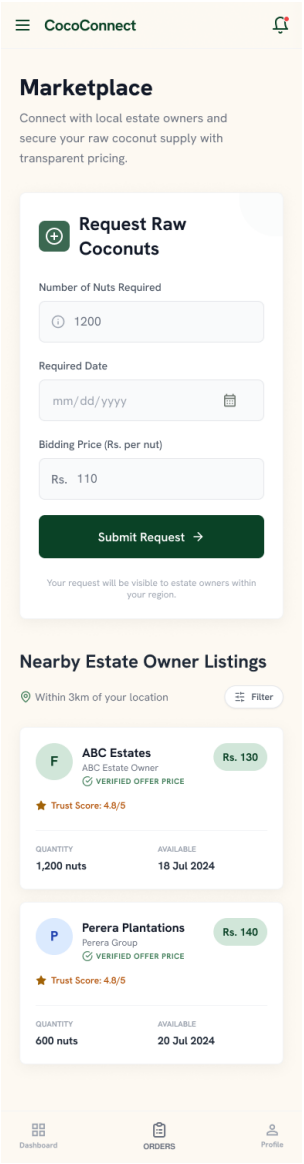


Figure 3: Marketplace – raw coconut request and nearby listings

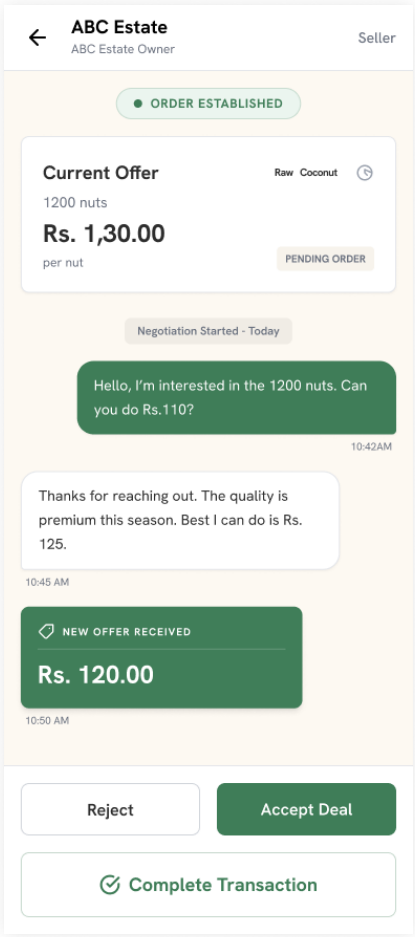


Figure 4: Negotiation

8. Design System

8.1 Colours

Swatch	Name	Hex	Used for
	Primary green	#2E7D32	App name, main buttons, selected tab, icons
	Dark green	#14432A	Strong buttons such as Submit Request
	Coconut brown	#B26A2C	Headings ('Buying'), offer highlights
	Cream	#FDF9F1	Screen background
	White	#FFFFFF	Cards
	Field fill	#F8F6F1	Text field background
	Border	#EDE6DA	Card and field borders
	Grey	#777777	Secondary text, inactive tabs

8.2 Typography

- One sans-serif family throughout.

- Large bold page titles; a two-tone title on the dashboard ('Buying' in brown, 'Categories' in light brown).
- Bold card titles, grey secondary text, prices in bold so they stand out.

8.3 Components

Component	Design
Category card	White, rounded corners, soft shadow, round icon badge, title and subtitle
Listing card	Avatar or photo, estate name, verified badge, trust score, price pill, quantity and date
Text field	Light fill, rounded, label above, hint inside
Primary button	Full width, rounded, green, white text with an arrow or icon
Offer card	Green card inside the chat with the offered price; Accept and Reject below
Bottom bar	Three items with icons and labels; selected item in green

9. Consistency with Functional Requirements

Requirement (SRS v3)	Screen that supports it
FR-04, FR-05 update profile and picture	Profile, Edit Profile
FR-07 log out	Profile
FR-10 browse products	Buying Categories, Coconut Categories, Marketplace
FR-11 search and filter by category, location and price	Coconut Categories, Marketplace filter
FR-12 view product details	Estate Owner Request
FR-13 price, quantity, method, location shown	Listing cards, Estate Owner Request
FR-14 product categories	Buying Categories, Coconut Categories
FR-15 contact sellers	Negotiate button, Negotiation
FR-37 display ratings	Trust score on listings, Profile
NFR-12 to NFR-14 usability	Consistent bottom bar, clear labels, inline validation messages